

Session 1

Task 1:

```
import java.io.*;

import java.util.*;

interface PerformOperation {

    boolean check(int a);

}

class MyMath {

    public static boolean checker(PerformOperation p, int num) {

        return p.check(num);

    }

    public PerformOperation isOdd() {

        return (int n) -> n % 2 != 0;

    }

    // 2. Check Prime / Composite

    public PerformOperation isPrime() {

        return (int n) -> {

            if (n < 2)

                return false;

            for (int i = 2; i * i <= n; i++) {

                if (n % i == 0)

                    return false;

            }

            return true;

        };

    }

};
```

```
}
```

```
// 3. Check Palindrome
```

```
public PerformOperation isPalindrome() {
```

```
    return (int n) -> {
```

```
        int original = n;
```

```
        int reverse = 0;
```

```
        while (n > 0) {
```

```
            int digit = n % 10;
```

```
            reverse = reverse * 10 + digit;
```

```
            n = n / 10;
```

```
        }
```

```
        return original == reverse;
```

```
    };
```

```
}
```

```
}
```

```
public class Solution {
```

```
    public static void main(String[] args) throws IOException {
```

```
        MyMath ob = new MyMath();
```

```
        BufferedReader br = new BufferedReader(new InputStreamReader(System.in));
```

```
        int T = Integer.parseInt(br.readLine());
```

```
        PerformOperation op;
```

```
        boolean ret = false;
```

```
        String ans = null;
```

```
        while (T--> 0) {
```

```
String s = br.readLine().trim();

StringTokenizer st = new StringTokenizer(s);

int ch = Integer.parseInt(st.nextToken());

int num = Integer.parseInt(st.nextToken());

if (ch == 1) {

    op = ob.isOdd();

    ret = ob.checker(op, num);

    ans = (ret) ? "ODD" : "EVEN";

} else if (ch == 2) {

    op = ob.isPrime();

    ret = ob.checker(op, num);

    ans = (ret) ? "PRIME" : "COMPOSITE";

} else if (ch == 3) {

    op = ob.isPalindrome();

    ret = ob.checker(op, num);

    ans = (ret) ? "PALINDROME" : "NOT PALINDROME";

}

System.out.println(ans);

}

}

}
```

Congratulations

You solved this challenge. Would you like to challenge your friends?

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✓ Test case 0

✓ Test case 1

✓ Test case 2

Compiler message

Success

Input (stdin)

1	5
2	1 4
3	2 5
4	3 898
5	1 3

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Task 2:

```
import java.util.*;
```

```
class Checker implements Comparator<Player> {
```

```
    @Override
```

```
    public int compare(Player a, Player b) {
```

```
        // Decreasing order of score
```

```
        if (a.score != b.score) {
```

```
            return b.score - a.score;
```

```
        }
```

```
        // Alphabetical order if scores are equal
```

```
        return a.name.compareTo(b.name);
```

```
    }
```

```
}
```

```
class Player{
```

String name;

int score;

Player(String name, int score){

 this.name = name;

 this.score = score;

}

}

class Solution {

 public static void main(String[] args) {

 Scanner scan = new Scanner(System.in);

 int n = scan.nextInt();

 Player[] player = new Player[n];

 Checker checker = new Checker();

 for(int i = 0; i < n; i++){

 player[i] = new Player(scan.next(), scan.nextInt());

 }

 scan.close();

 Arrays.sort(player, checker);

 for(int i = 0; i < player.length; i++){

 System.out.printf("%s %s\n", player[i].name, player[i].score);

 }

 }

}import java.util.*;

```
class Checker implements Comparator<Player> {
```

```
    @Override
```

```
    public int compare(Player a, Player b) {
```

```
        // Decreasing order of score
```

```
        if (a.score != b.score) {
```

```
            return b.score - a.score;
```

```
        }
```

```
        // Alphabetical order if scores are equal
```

```
        return a.name.compareTo(b.name);
```

```
    }
```

```
}
```

```
class Player{
```

```
    String name;
```

```
    int score;
```

```
    Player(String name, int score){
```

```
        this.name = name;
```

```
        this.score = score;
```

```
    }
```

```
}
```

```
class Solution {
```

```
    public static void main(String[] args) {
```

```
        Scanner scan = new Scanner(System.in);
```

```
        int n = scan.nextInt();
```

```

Player[] player = new Player[n];

Checker checker = new Checker();

for(int i = 0; i < n; i++){

    player[i] = new Player(scan.next(), scan.nextInt());

}

scan.close();

Arrays.sort(player, checker);

for(int i = 0; i < player.length; i++){

    System.out.printf("%s %s\n", player[i].name, player[i].score);

}

}

}

```

Congratulations

You solved this challenge. Would you like to challenge your friends? [Twitter](#) [LinkedIn](#)

[Next Challenge](#)

✓ Test case 0

✓ Test case 1 

✓ Test case 2 

✓ Test case 3 

✓ Test case 4 

✓ Test case 5 

Input (stdin)

```

1 5
2 amy 100
3 david 100
4 heraldo 50
5 aakansha 75
6 aleksa 150

```

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Expected Output

```

1 aleksa 150
2 amy 100
3 david 100

```

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